Figure 10.27 An example of the different partitions produced by the split heuristic in the original Ss-tree version (center), and the k-means split heuristic (left). For the original heuristic, the direction of maximum variance was along the y axis. For this example, we assume `M==12` and `m==6`.

Although newer, more complex clustering algorithms have been developed during the years (as we'll see in chapter 12, where we'll also describe DBSCAN and Optics), k-means is still a perfect match for Ss-trees, because it naturally produces spherical clusters, each with a centroid equal to the center of mass of the points in the cluster.

10.5.4 Reducing Overlap

The k-means-split heuristic is a powerful tool to reduce node overlapping and keep the tree balanced and search fast; as we have already reminded at the beginning of last section, however, we can unbalance a node also while deleting points, in particular during merge or when we move a point/child across siblings. Moreover, sometimes the overlapping can be the result of several operations on the tree and involve more than just two nodes, or even more than a single level.

Finally, the k-means-split heuristic itself, not having overlap minimization as a criterion, and because of the intrinsic behavior of k-means, could produce results where a node with larger variance might completely overlap a smaller node.